Luke Guerdan

✉ lukemguerdan@gmail.com
lukeguerdan.com

Experience

2026–Current **Duolingo**, *AI Research Engineer II*.

Develop evaluation methods and infrastructure for AI-enabled learning experiences as a member of Duolingo's AI Platform team.

2021–2026 **Carnegie Mellon University**, *Ph.D. Researcher*.

Advisors: Kenneth Holstein, Steven Wu.

Developed statistical methods and sociotechnical frameworks for evaluating the capabilities, risks, and limitations of AI systems. First-author research published at FAccT (2×), ICML, ICLR, NeurIPS, and CSCW, including a Best Paper Award at FAccT 2023.

2024 **Microsoft Research NYC**, *Ph.D. Research Intern*, Fairness, Accountability, Transparency & Ethics (FATE) Team.

Mentors: Alexandra Chouldechova, Solon Barocas, Hanna Wallach.

Designed a statistical framework for validating LLM-as-a-judge systems on subjective evaluation tasks, working closely with applied scientists to ground the methodology in practical needs. Identified four actionable implications for LLM-as-a-judge evaluation design, some of which were adopted internally. Published at NeurIPS '25 along with public code, data, and blog post.

2018–2021 **TigerAware, LLC**, *Co-founder*.

Co-founded TigerAware, a full-stack mobile data collection platform for addiction research. Designed the system architecture and led a six-person technical team. Deployed the platform across 18+ U.S. research institutions, supporting 15,000+ research participants and processing millions of participant responses. Negotiated multi-year service contracts with institutional legal and compliance teams. Platform remains in active use [web, press].

2019 **Washington University in St. Louis**, *Summer Research Intern*, NSF REU in Big Data Analytics.

Advisor: Chien-Ju Ho

Investigated primal-dual based techniques for efficient online allocation of scarce societal resources.

2015–2019 **University of Missouri**, *Research Assistant*.

Investigated machine learning techniques for understanding human physiology and behavior.

Education

2021-2026 **Carnegie Mellon University**.

Ph.D. in Human-Computer Interaction

Advisors: Steven Wu, Kenneth Holstein

Thesis: Principled Measurement and Evaluation of AI Systems under Practical Constraints

2020–2021 **Cambridge University**.

MPhil in Advanced Computer Science (Honors with Distinction)

2015–2019 **University of Missouri**.

B.S. in Computer Science, Psychology (GPA: 4.0/4.0)

Selected Academic Publications

Google Scholar Link *(An asterisk (*) below indicates joint first-author.)*

2026 **Luke Guerdan***, Justin Whitehouse*, Kimberly Truong*, Kenneth Holstein, and Zhiwei Steven Wu. Doubly-robust LLM-as-a-judge: Externally valid estimation with imperfect personas. *Proceedings of the International Conference on Learning Representations (ICLR '26)*, 2026.

2026 **Luke Guerdan**, Jessica Huynh, Kiruthika Raja, Razvan Amironesei, Jeffrey Bigham, Fernando Diaz, Craig Greenberg, Patrick Hall, Renee Shelby, Zhiwei Steven Wu, and Kenneth Holstein. RubricStudio: Operationalizing construct validity principles in cross-functional generative ai evaluation design. *In preparation*, 2026.

2025 Lydia T Liu*, Inioluwa Deborah Raji*, Angela Zhou*, **Luke Guerdan**, Jessica Hullman, Daniel Malinsky, Bryan Wilder, Simone Zhang, Hammaad Adam, Amanda Coston, and others. Bridging prediction and intervention problems in social systems. *arXiv preprint arXiv:2507.05216*, 2025.

2025 **Luke Guerdan***, Devansh Saxena*, Stevie Chancellor, Zhiwei Steven Wu, and Kenneth Holstein. Measurement as bricolage: Examining how data scientists construct target variables for predictive modeling tasks. In *Proceedings of the 28th ACM SIGCHI Conference on Computer-Supported Cooperative Work and Social Computing (CSCW '25)*, 2025.

2025 **Luke Guerdan**, Solon Barocas, Kenneth Holstein, Hanna Wallach, Zhiwei Steven Wu, and Alexandra Chouldechova. Validating LLM-as-a-judge systems under rating indeterminacy. In *Advances in Neural Information Processing Systems (NeurIPS '25)*, 2025.

2024 **Luke Guerdan**, Amanda Coston, Kenneth Holstein, and Zhiwei Steven Wu. Predictive performance comparison of decision policies under confounding. In *Proceedings of the International Conference on Machine Learning (ICML '24)*, 2024.

2023 Anna Kawakami, **Luke Guerdan**, Yang Cheng, Kate Glazko, Matthew Lee, Scott Carter, Nikos Arechiga, Haiyi Zhu, and Kenneth Holstein. Training towards critical use: Learning to situate AI predictions relative to human knowledge. In *Conference on Collective Intelligence (CI '23)*, 2023.

2023 **Luke Guerdan**, Amanda Coston, Zhiwei Steven Wu, and Kenneth Holstein. Ground(less) truth: A causal framework for proxy labels in human-algorithm decision-making. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*, 2023.

2023 **Luke Guerdan**, Amanda Coston, Kenneth Holstein, and Zhiwei Steven Wu. Counterfactual prediction under outcome measurement error. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*, 2023. Best Paper Award.

Technical Skills

Languages: Python, R, C/C++, JavaScript, SQL

Libraries: PyTorch, Pandas, NumPy, SciPy

Tools: UNIX, Git, Jupyter

Awards & Honors

2023 Best Paper Award at ACM Conference on Fairness, Accountability, and Transparency (FAccT)

2023 PhD Fellow, Center for Advancing Safety of Machine Intelligence (CASMI) at Northwestern University

2020 National Science Foundation Graduate Research Fellowship

2019 University of Missouri Award for Academic Distinction

2018 DAAD RISE Fellowship

2018 Barry Goldwater Scholarship in Science and Engineering

Selected Talks & Speaking Engagements

2026 **Invited Speaker**. Simons Institute Workshop on Bridging Predictions and Interventions in Social Systems. *'Measurement Interventions for AI Design and Evaluation.'*

Invited Speaker. Abridge Reading Group on Automated Evaluation in Healthcare. *'Validating LLM-as-a-Judge Systems under Rating Indeterminacy.'*

2025 **Invited Speaker**. FAR.AI San Diego Alignment Workshop. *'Validating LLM-as-a-Judge Systems under Rating Indeterminacy.'*

Invited Speaker. Evaluating Evaluations Workshop at NeurIPS 2025. *'RubricStudio: Operationalizing Construct Validity Principles in Cross Functional Generative AI Evaluation Design.'*

Invited Speaker. INFORMS 2025 Session on Human-AI Collaboration for Decision-Making. *'Validating LLM-as-a-Judge Systems under Rating Indeterminacy.'*

Invited Speaker. Pareto AI Seminar Series. *'Principled Measurement and Evaluation of AI Systems Under Practical Constraints.'*

Invited Speaker. Amazon Science Responsible AI Seminar. *'Validating LLM-as-a-Judge Systems in the Absence of Gold Labels.'*

2024 **Invited Speaker**. MILA/McGill NLP Reading Group. Virtual. *'Towards Principled Evaluation Under Imperfect "Ground Truth" Labels.'*

Invited Speaker. INFORMS 2024 Session on Human-Centered AI and Decision Making for Social Good. Seattle, WA. *'A Methodological Framework for Human-Algorithm Performance Comparisons Under Incomplete Information.'*

Invited Speaker. BIRS Workshop on Bridging Prediction and Intervention Problems in Social Systems. Banff International Research Station, Calgary CA. *'A Framework for Human-Algorithm Performance Comparison Under Incomplete Information.'*

Invited Speaker. Complexity Science Hub Vienna, Lecture Series on Network Inequality. Remote. *'Human-Algorithm Decision-Making Under Imperfect Proxy Labels.'*

2023 **Podcast Guest**. Still Updating Podcast. Remote. *'Finding Truth in Labeled Data.'*

Service

Leadership.

2025 Lead Organizer. CMU AI Measurement Science and Engineering Center (AIMSEC) Webinar Series. Organized webinar series on measurement and evaluation of AI systems. Invited speakers, promoted online events, and moderated panels. 200+ participants attended across three panels.

2023–2024 Lead Organizer. CMU Fairness, Accountability, Transparency, and Ethics Reading Group. Organized weekly reading group discussing research on societal impacts of AI systems. Grew attendance to 20-40 weekly participants through external speakers and student research talks.

Reviewing (PC).

2025 ICLR, NeurIPS, FAccT, CHI, ACM Journal on Responsible Computing

2024 ICLR, FAccT, DIS, CHI

2023 ICLR, AAAI, CHI, FAccT, ACM Journal on Responsible Computing

Mentorship

2025 Kate Lee, Master's Student, Summer Research Assistant.

2023 Ming Wang, High School Student, Summer Research Internship.

2022 Mahika Varma, Undergraduate, HCI Independent Study.

2022 Angelica Bonilla, Undergraduate, CMU AI Mentorship Program.

2021 Matthew Ok, Undergraduate, HCI Independent Study.